Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Kieffer, John Ph.D.		
Project Title:	Structure and Dynamics of Fragile Glass-Forming Systems		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	05/01/2003	**End Date:**	10/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Ray, Chandra	**Contact Phone:**	256 544-2918
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNM04AA14G		
Performance Goal No.:			
Performance Goal Text:			

This project consists of the investigation of a series of fragile glass formers, including phosphates, chalcogenides, and aluminates, with the aim to elucidate structural developments associated with glass formation. To this end we use a combination of concurrent Raman and Brillouin light scattering as the structural probe, and molecular dynamics simulations to interpret the experimental observations. Raman scattering allows us to monitor changes in the symmetry of structural building blocks, while Brillouin scattering yields the complex mechanical modulus at the molecular scale. The storage component of this modulus provides a measure of the degree of connectivity between building blocks and the structural integrity, whereas the loss modulus assesses the energy dissipated in aperiodic motions of these building blocks. The role of simulation is to explain the observed measures in terms of structural changes that occur during glass formation.
This methodology constitutes a new approach towards investigating glass structure and dynamics. In particular, it

Task Description:	provides insights into the elusive intermediate-range order of non-crystalline structures. Variable parameters in this study include the nature of the network former and the chemical composition of the glass. In select cases we carry out a comparative study of glasses prepared via sol-gel synthesis and melt processing is envisaged. This project has a three-tiered objective: 1. Devise and refine a methodology, based on concurrent Brillouin and Raman scattering, to characterize structural developments in fluids and supercooled melts during the glass formation process. While we have already demonstrated that the scattering techniques can act as a structural probe on the basis of comparative evaluation, we expect that the combination with molecular dynamics simulations will allow for a direct mechanistic and structural interpretation of experimental observables. 2. Substantiate and elucidate the concept of polyamorphism, i.e., the existence of more than one thermodynamically and structurally distinct amorphous state for a given material. Specifically, we will verify whether polyamorphic transitions are ubiquitous to glass formation. 3. By understanding the fundamental principles that underlie glass formation and that control structural developments in glass-forming systems, develop knowledge that allows for a more targeted design of amorphous materials with specific properties. With this project we take a novel approach towards investigating the amorphous state of matter, and therefore expect to gain unprecedented insights in this subject matter. Accordingly, we expect to (i) advance fundamental aspects of glass science, (ii) devise knowledge for the creation of new glass-forming materials with applications in optical waveguides, photonic crystals, optical amplifiers, lasers. sensors, integrated circuits, and logic devices, and (iii) develop a technique of investigation that is of general interest for NASA's microgravity research program because it allows for the contact-free measurement of visco-elastic properties of fluids, i.e., without obliterating the environment free of external forces provided by microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While ultimately container-less melting in microgravity conditions provides the ideal environment for investigating structural developments in deeply supercooled glass-forming melts or in a sol-gel precursor maintained in delicate thermo-mechanical equilibrium, the method of investigation developed in this research can be applied in numerous situations on earth in which direct contact between specimen and the mechanical probe is difficult or impossible. Such situations include specimens that are (i) chemically aggressive, (ii) held at very high temperature, (iii) of fragile constitution, (iv) of small dimensions, and (v) part of a living organism.
Task Progress:	While the grant for this research was awarded at the end of 2003, a graduate student qualified to work on the project could not be identified until September 2004. The project is underway at full capacity since that time. Prior to September 2004, i.e., in the spring and summer of that year, the PI and a part-time postdoctoral research associate conducted preparatory work for the project. This consisted of participating in the design refinements of the Fabry-Perot interferometer, which is used for Brillouin spectroscopy and is manufactured by JSR Scientific instruments in Switzerland. The newly designed cosine interferometer is more robust and characterized by higher operating stability. Another important feature of this apparatus is that it can be outfitted with a confocal interferometer stage, which allows for higher-resolution measurements. At this time, the start-up phase of the experimental aspect of the project is nearly completed. The interferometer has been received in mid September. It has been set up in our laboratory, aligned, and calibrated. Meanwhile the graduate student working on the project is being trained in using the instrument. He has also familiarized himself with sample preparation techniques and is now working to prepare chalcogenide glasses. Once we have glasses of satisfactory quality, we will develop high-temperature sample stages to observe structural evolutions in these materials through the glass transition range. The part-time postdoc, although mostly supported by a different grant, has developed simulation code that will be used in this project. In particular, she enhanced our existing reactive force field to allow for multiple coordination states of the network-forming element. Based on her simulations, we have been able to explain the thermo-mechanical anomalies in silica glass, i.e., the increase of the elastic modulus with temperature and the decrease of the modulus with pressure. In fact, we can attribute these anomalous behaviors to polyamorphic transitions. This is also the reason for the irreversible densification of silica glass at high pressure. In earlier work we discovered a reversible polyamorphic transition in boron oxide glass under pressure, which has unusual characteristics in that it is continuous on compression but discontinuous on decompression. MD simulations confirm that this transition involves the changeover between three- and four-coordinated states for boron.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
Articles in Peer-reviewed Journals	Huang L, Kieffer J. "Amorphous-amorphous transitions in silica glass. I. Reversible transitions and thermomechanical anomalies." Physical Review B 2004 Jun;69(22):224203. http://dx.doi.org/10.1103/PhysRevB.69.224203 , Jun-2004
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Articles in Peer-reviewed Journals	Nicholas JD, Sinogeikin SV, Kieffer J, Bass JD. "Spectroscopic evidence of polymorphism in vitreous B2O3." Physical Review Letters 2004 May;92(21):215701. http://dx.doi.org/10.1103/PhysRevLett.92.215701 , May-2004

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